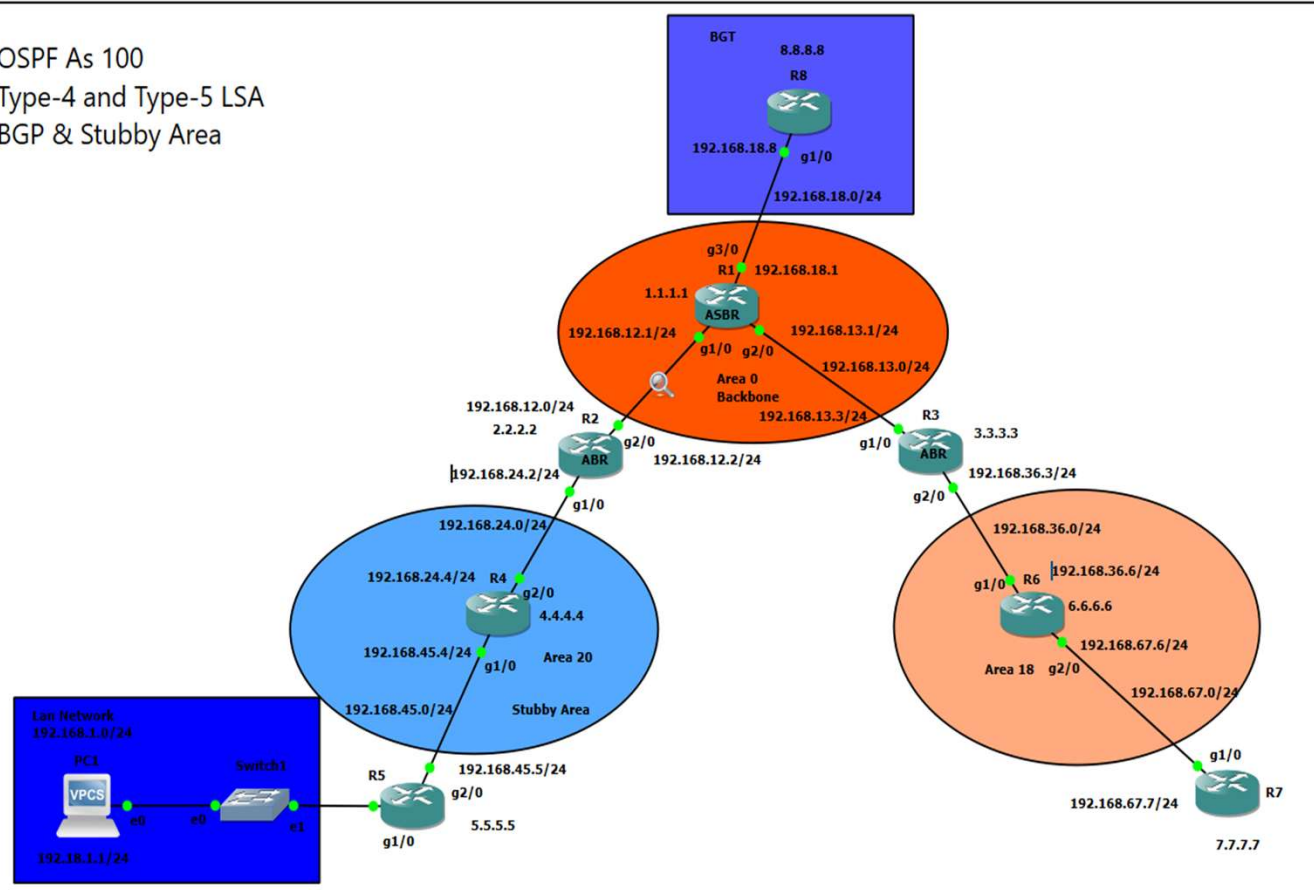




OSPF As 100
Type-4 and Type-5 LSA
BGP & Stubby Area



Source OSPF Router: 1.1.1.1
 Area ID: 0.0.0.0 (Backbone)
 Checksum: 0x683e [correct]
 Instance ID: Base IPv4 Unicast Instance (0)
 Auth Type: Null (0)
 Auth Data (none): 0000000000000000

▼ LS Update Packet

Number of LSAs: 1

▼ LSA-type 1 (Router-LSA), len 48

.000 0000 0000 0010 = LS Age (seconds): 2

0... = Do Not Age Flag: 0

▼ Options: 0x22, (DC) Demand Circuits, (E) External Routing

0... = DN: Not set

.0.. = (O) Opaque: Not set

..1. = (DC) Demand Circuits: Supported

...0 = (L) LLS Data block: Not Present

.... 0... = (N) NSSA: Not supported

.... .0.. = (MC) Multicast: Not capable

.... ..1. = (E) External Routing: Capable

.... ...0 = (MT) Multi-Topology Routing: No

LS Type: Router-LSA (1)

Link State ID: 3.3.3.3

Advertising Router: 3.3.3.3

Sequence Number: 0x80000008

Checksum: 0x3dc7

Length: 48

▼ Flags: 0x01, (B) Area border router

0... = (H) Host: No

..0. = (S) Shortcut-capable ABR: No

...0 = (N) NSSA translation: No

.... 0... = (W) Wild-card multicast receiver: No

.... .0.. = (V) Virtual link endpoint: No

.... ..0. = (E) AS boundary router: No

.... ...1 = (B) Area border router: Yes

Number of Links: 2

> Type: Stub ID: 3.3.3.3 Data: 255.255.255.255 Metric: 1

> Type: Transit ID: 192.168.13.1 Data: 192.168.13.3 Metric: 1

Router 2 announced as ABR router

B=1

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 8.8.8.8, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 108/121/132 ms
R5(config)#do show ip ospf database
```

OSPF Router with ID (5.5.5.5) (Process ID

Router Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	172	0x80000007	0x0047CD	1
4.4.4.4	4.4.4.4	205	0x80000009	0x00CAD0	3
5.5.5.5	5.5.5.5	1679	0x80000007	0x006FBE	3

Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.24.2	2.2.2.2	172	0x80000005	0x0099F8
192.168.45.4	4.4.4.4	205	0x80000004	0x00D98E

Summary Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
1.1.1.1	2.2.2.2	673	0x80000005	0x002BFF
2.2.2.2	2.2.2.2	424	0x80000004	0x00F434
3.3.3.3	2.2.2.2	1446	0x80000003	0x00DC47
6.6.6.6	2.2.2.2	1446	0x80000002	0x005EB9
7.7.7.7	2.2.2.2	1446	0x80000002	0x003AD8
192.168.12.0	2.2.2.2	673	0x80000005	0x0019A1
192.168.13.0	2.2.2.2	673	0x80000005	0x0018A0
192.168.18.0	2.2.2.2	673	0x80000005	0x00E0D2
192.168.36.0	2.2.2.2	1446	0x80000002	0x002A79
192.168.67.0	2.2.2.2	1446	0x80000002	0x00DDA5

Summary ASB Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
1.1.1.1	2.2.2.2	172	0x80000002	0x001915

Type-5 AS External Link States

Link ID	ADV Router	Age	Seq#	Checksum	Tag
8.8.8.8	1.1.1.1	303	0x80000001	0x0043FF	200

Router-5

We can see type-5,4 LSA

```
R5
R4
R2
R5(config)#
R5(config)#
R5(config)#
R5(config)#router ospf 100
R5(config-router)#area 20 stub
R5(config-router)#
*Sep 11 17:52:22.123: %OSPF-5-ADJCHG: Process 100, Nbr 4.4.4.4 on GigabitEthernet2/0 from FULL to DOWN, Neighbor Down: Adj
jacency forced to reset
R5(config-router)#
*Sep 11 17:52:58.735: %OSPF-5-ADJCHG: Process 100, Nbr 4.4.4.4 on GigabitEthernet2/0 from LOADING to FULL, Loading Done
R5(config-router)#do show ospf database
R5(config-router)#do show ip ospf database
```

OSPF Router with ID (5.5.5.5) (Process ID 100)

Router Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	81	0x80000009	0x00759D	1
4.4.4.4	4.4.4.4	81	0x8000000C	0x003166	3
5.5.5.5	5.5.5.5	150	0x8000000A	0x009D8E	3

Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.24.4	4.4.4.4	81	0x80000001	0x004F3F
192.168.45.5	5.5.5.5	150	0x80000001	0x00C5A2

Summary Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
0.0.0.0	2.2.2.2	85	0x80000001	0x0075C0
1.1.1.1	2.2.2.2	85	0x80000006	0x0047E4
2.2.2.2	2.2.2.2	85	0x80000005	0x001119
3.3.3.3	2.2.2.2	85	0x80000005	0x00F62D
6.6.6.6	2.2.2.2	85	0x80000004	0x00789F
7.7.7.7	2.2.2.2	85	0x80000004	0x00548E
192.168.12.0	2.2.2.2	85	0x80000006	0x003586
192.168.13.0	2.2.2.2	85	0x80000006	0x003485
192.168.18.0	2.2.2.2	85	0x80000006	0x00FCB7
192.168.36.0	2.2.2.2	85	0x80000004	0x00445F
192.168.67.0	2.2.2.2	85	0x80000004	0x00F78B

```
R5(config-router)#
R5(config-router)#
R5(config-router)#
R5(config-router)#
```

In Router-5

We can see after configure
Stub as area 20.

Stub Area stops Type-4 LSA

Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.24.4	4.4.4.4	81	0x80000001	0x004F3F
192.168.45.5	5.5.5.5	150	0x80000001	0x00C5A2

Summary Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
0.0.0.0	2.2.2.2	85	0x80000001	0x0075C0
1.1.1.1	2.2.2.2	85	0x80000006	0x0047E4
2.2.2.2	2.2.2.2	85	0x80000005	0x001119
3.3.3.3	2.2.2.2	85	0x80000005	0x00F62D
6.6.6.6	2.2.2.2	85	0x80000004	0x00789F
7.7.7.7	2.2.2.2	85	0x80000004	0x0054BE
192.168.12.0	2.2.2.2	85	0x80000006	0x003586
192.168.13.0	2.2.2.2	85	0x80000006	0x003485
192.168.18.0	2.2.2.2	85	0x80000006	0x00FCB7
192.168.36.0	2.2.2.2	85	0x80000004	0x00445F
192.168.67.0	2.2.2.2	85	0x80000004	0x00F78B

```

R5(config-router)#
R5(config-router)#
R5(config-router)#
R5(config-router)#area 20 stub no-summary
R5(config-router)#do show ip ospf database

```

OSPF Router with ID (5.5.5.5) (Process ID 100)

Router Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
2.2.2.2	2.2.2.2	583	0x80000009	0x00759D	1
4.4.4.4	4.4.4.4	583	0x8000000C	0x003166	3
5.5.5.5	5.5.5.5	652	0x8000000A	0x009D8E	3

Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
192.168.24.4	4.4.4.4	582	0x80000001	0x004F3F
192.168.45.5	5.5.5.5	652	0x80000001	0x00C5A2

Summary Net Link States (Area 20)

Link ID	ADV Router	Age	Seq#	Checksum
0.0.0.0	2.2.2.2	8	0x80000003	0x0071C2

R5(config-router)#

In router we configured stub no-summary , then
 Type-3 LSA also stops in OSPF Database
 By this OSPF Features we can save minimize Database